

EE 3TR4 (COMMUNICATION SYSTEMS): QUIZ 1

INSTRUCTOR: DR. JUN CHEN

Student Name: _____

Student Number: _____

Compute the Fourier transform $X(f)$ of the following signal:

$$x(t) = \text{sinc}(100t)\text{sinc}(150t).$$

Fourier transform of special functions:

$$\text{rect}(t) \xleftrightarrow{\mathcal{F}} \text{sinc}(f),$$

$$\text{triang}(t) \xleftrightarrow{\mathcal{F}} \text{sinc}^2(f),$$

where

$$\text{rect}(t) = \begin{cases} 1, & |t| < \frac{1}{2}, \\ 0, & \text{otherwise}, \end{cases}$$

$$\text{triang}(t) = \begin{cases} 1 - |t|, & |t| < 1, \\ 0, & \text{otherwise}, \end{cases}$$

$$\text{sinc}(t) = \frac{\sin(\pi t)}{\pi t}.$$

Properties of Fourier transform:

If $g(t) \xleftrightarrow{\mathcal{F}} G(f)$ and $h(t) \xleftrightarrow{\mathcal{F}} H(f)$, then

$$g(at) \xleftrightarrow{\mathcal{F}} \frac{1}{|a|} G\left(\frac{f}{a}\right),$$

$$G(t) \xleftrightarrow{\mathcal{F}} g(-f),$$

$$g(t - t_0) \xleftrightarrow{\mathcal{F}} G(f) e^{-j2\pi f t_0},$$

$$g(t) e^{j2\pi f_c t} \xleftrightarrow{\mathcal{F}} G(f - f_c),$$

$$g(t) \cos(2\pi f_c t) \xleftrightarrow{\mathcal{F}} \frac{1}{2} [G(f - f_c) + G(f + f_c)],$$

$$g(t) * h(t) \xleftrightarrow{\mathcal{F}} G(f) H(f),$$

$$g(t) h(t) \xleftrightarrow{\mathcal{F}} G(f) * H(f).$$

Solution to Quiz 1

Using the fact that $g(t)h(t) \xleftrightarrow{\mathcal{F}} G(f) * H(f)$, $g(at) \xleftrightarrow{\mathcal{F}} \frac{1}{|a|}G\left(\frac{f}{a}\right)$, $\text{rect}(t) \xleftrightarrow{\mathcal{F}} \text{sinc}(f)$, and $G(t) \xleftrightarrow{\mathcal{F}} g(-f)$, we have

$$\begin{aligned} X(f) &= \mathcal{F}[\text{sinc}(100t)\text{sinc}(150t)] \\ &= \mathcal{F}[\text{sinc}(100t)] * \mathcal{F}[\text{sinc}(150t)] \\ &= \left(\frac{1}{100} \text{rect}\left(\frac{f}{100}\right) \right) * \left(\frac{1}{150} \text{rect}\left(\frac{f}{150}\right) \right) \\ &= \begin{cases} \frac{1}{150}, & |f| \leq 25, \\ \frac{125-|f|}{15000}, & 25 < |f| \leq 125, \\ 0, & |f| > 125. \end{cases} \end{aligned}$$

EE 3TR4 C02 (COMMUNICATION SYSTEMS): QUIZ 1

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Find and sketch the Fourier transform, $X(f)$, of the following signal:

$$x(t) = 10\text{sinc}(50t)\text{sinc}(100t).$$

Fourier transform of special functions:

$$\text{rect}(t) \xleftrightarrow{\mathcal{F}} \text{sinc}(f),$$

$$\text{triang}(t) \xleftrightarrow{\mathcal{F}} \text{sinc}^2(f),$$

where

$$\text{rect}(t) = \begin{cases} 1, & |t| < \frac{1}{2}, \\ 0, & \text{otherwise}, \end{cases}$$

$$\text{triang}(t) = \begin{cases} 1 - |t|, & |t| < 1, \\ 0, & \text{otherwise}, \end{cases}$$

$$\text{sinc}(t) = \frac{\sin(\pi t)}{\pi t}.$$

Properties of Fourier transform:

If $g(t) \xleftrightarrow{\mathcal{F}} G(f)$ and $h(t) \xleftrightarrow{\mathcal{F}} H(f)$, then

$$g(at) \xleftrightarrow{\mathcal{F}} \frac{1}{|a|} G\left(\frac{f}{a}\right),$$

$$G(t) \xleftrightarrow{\mathcal{F}} g(-f),$$

$$g(t - t_0) \xleftrightarrow{\mathcal{F}} G(f) e^{-j2\pi f t_0},$$

$$g(t) e^{j2\pi f_c t} \xleftrightarrow{\mathcal{F}} G(f - f_c),$$

$$g(t) \cos(2\pi f_c t) \xleftrightarrow{\mathcal{F}} \frac{1}{2} [G(f - f_c) + G(f + f_c)],$$

$$g(t) * h(t) \xleftrightarrow{\mathcal{F}} G(f) H(f),$$

$$g(t) h(t) \xleftrightarrow{\mathcal{F}} G(f) * H(f).$$

Solution to Quiz 1

Using the fact that $g(t)h(t) \xleftrightarrow{\mathcal{F}} G(f) * H(f)$, $g(at) \xleftrightarrow{\mathcal{F}} \frac{1}{|a|}G\left(\frac{f}{a}\right)$, $\text{rect}(t) \xleftrightarrow{\mathcal{F}} \text{sinc}(f)$, and $G(t) \xleftrightarrow{\mathcal{F}} g(-f)$, we have

$$\begin{aligned}
 X(f) &= \mathcal{F}[10\text{sinc}(50t)\text{sinc}(100t)] \\
 &= 10\mathcal{F}[\text{sinc}(50t)\text{sinc}(100t)] \\
 &= 10\mathcal{F}[\text{sinc}(50t)] * \mathcal{F}[\text{sinc}(100t)] \\
 &= 10\left(\frac{1}{50}\text{rect}\left(\frac{f}{50}\right)\right) * \left(\frac{1}{100}\text{rect}\left(\frac{f}{100}\right)\right) \\
 &= \frac{1}{500}\left(\text{rect}\left(\frac{f}{50}\right)\right) * \left(\text{rect}\left(\frac{f}{100}\right)\right) \\
 &= \begin{cases} 0, & f < -75, \\ \frac{t+75}{500}, & -75 \leq f \leq -25, \\ \frac{1}{10}, & -25 \leq f \leq 25, \\ \frac{-t+75}{500}, & 25 \leq f \leq 75, \\ 0, & f > 75. \end{cases}
 \end{aligned}$$

